Closing the $n = 6$ Base Case for the Second q-Shifted Pair: Polynomial Identity via Degree Bound and Multi-Point Evaluation

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Abstract

The second q-shifted pair theorem (companion paper, `2026-04-19-second-q-shifted-pair.tex`) reduces, by locality, to the $n = 6$ base identity

$$F(q) \;:=\; \det M^{(5)}[D_6] \cdot \det M^{(3)}[D_6] \;\overset{?}{=}\; q^{90}\big(\det M^{(4)}[D_6]\big)^2 \;=:\; G(q) \quad \text{in } \mathbb{Z}[q].$$

The companion paper verifies this identity at ten distinct primes $q \in \{2, 3, 5, 7, 11, 13, 17, 19, 23, 29\}$ but does not establish it as a polynomial identity. Here we close the gap: we derive a tight *a priori* degree bound $\deg(F - G) \le 1444$ from the per-column q-degrees of $M^{(k)}[D_6]$, and we verify $F(q_0) = G(q_0)$ exactly at 1449 distinct positive integers $q_0 \in \{2, 3, \ldots, 1450\}$ via fraction-free (Bareiss) determinant evaluation. Since $1449 > 1444$, the polynomial $F - G$ vanishes at more distinct points than its degree allows, hence $F - G \equiv 0$ in $\mathbb{Z}[q]$.

1 Setup and statement

We use the notation of the companion paper. Briefly: $H_q(S_n)$ is the Iwahori–Hecke algebra with quadratic relation $T_i^2 = (q-1)T_i + q$; $\Pi_q^{(k)} := B_1 \cdots B_k$ with $B_j := (1+T_j)\cdots(1+T_1)$; $D_n := \{w \in S_n : w(2i-1) > w(2i)\ \forall\, 1 \le i \le \lfloor n/2 \rfloor\}$ of size $|D_n| = n!/2^{\lfloor n/2 \rfloor}$; and $M^{(k)}[u, v] := [T_u]\,(T_v \cdot \Pi_q^{(k)})$ for $u, v \in D_n$.

Theorem 1 (Base case at $n = 6$)**.**

$$\det M^{(5)}[D_6] \cdot \det M^{(3)}[D_6] \;=\; q^{90}\big(\det M^{(4)}[D_6]\big)^2 \quad \textit{in } \mathbb{Z}[q].$$

The remainder of this note proves Theorem 1.

2 Degree bound

Lemma 2. *For $u, v \in D_n$ and any $k \in \{1, \ldots, n-1\}$, the polynomial $M^{(k)}[u, v] \in \mathbb{Z}[q]$ has q-degree at most $k(k+1)/2$.*

Proof. The element $\Pi_q^{(k)} = B_1 \cdots B_k$ is a product of $\sum_{j=1}^{k} j = k(k+1)/2$ factors of the form $(1 + T_i)$. Write $x = \sum_w c_w(q)\, T_w \in H_q(S_n)$ with $c_w \in \mathbb{Z}[q]$ and let $d(x) := \max_w \deg c_w$. The Hecke quadratic relation gives, for each w and each simple reflection s_i,

$$T_w \cdot (1 + T_i) \;=\; \begin{cases} T_w + T_{ws_i}, & \text{if } \ell(ws_i) > \ell(w), \\ (1 + q - 1)\, T_w + q\, T_{ws_i} \;=\; q\, T_w + q\, T_{ws_i}, & \text{if } \ell(ws_i) < \ell(w). \end{cases}$$

In both cases each $\mathbb{Z}[q]$-coefficient of $x \cdot (1 + T_i)$ in the T-basis is a sum of products of an old coefficient with a polynomial of degree at most 1 (namely 1 or q), so $d(x \cdot (1 + T_i)) \leq d(x) + 1$. Starting with $x_0 = T_v$ (so $d(x_0) = 0$) and applying the $k(k+1)/2$ factors of $\Pi_q^{(k)}$ in turn yields $\deg M^{(k)}[u, v] \leq d(T_v \cdot \Pi_q^{(k)}) \leq k(k+1)/2$. □

Lemma 3 (Per-column degree analysis at $n = 6$)**.** *Let* $C_k := \sum_{v \in D_6} \max_{u \in D_6} \deg M^{(k)}[u, v]$ *be the sum over columns of the per-column maximum q-degree. Then*

$$C_3 = 405, \qquad C_4 = 668, \qquad C_5 = 1054.$$

In particular, $\deg \det M^{(3)}[D_6] \leq 405$*,* $\deg \det M^{(4)}[D_6] \leq 668$*, and* $\deg \det M^{(5)}[D_6] \leq 1054$*.*

Proof. The bound $\deg \det M \leq C_k$ follows from $\det M = \sum_{\sigma \in S_{|D_6|}} \operatorname{sgn}(\sigma) \prod_v M[\sigma(v), v]$, since for each σ, $\deg \prod_v M[\sigma(v), v] = \sum_v \deg M[\sigma(v), v] \leq \sum_v \max_u \deg M[u, v] = C_k$.

The exact values $C_3 = 405$, $C_4 = 668$, $C_5 = 1054$ are obtained by direct computation: build $M^{(k)}[D_6]$ as a 90×90 matrix over $\mathbb{Z}[q]$ via SymPy's symbolic Hecke right-action, then read off each entry's $\mathbb{Q}[q]$-degree. The script `/home/clio/projects/scratch/d5_D6_degree_analysis.py` reproduces these values; the per-column maximum degrees are independent of q and depend only on the combinatorial structure of D_6 and the staircase word. (The analogous row-max sums are 540, 840, and 1350, also valid upper bounds; we use the column variant since it is tighter.)

The closed form $\det M^{(3)}[D_6] = q^{270}(q+1)^{90}(2q+1)^{30}$ (by locality from $\det M^{(3)}[D_4] = q^{18}(q+1)^6(2q+1)^2$ raised to the $|D_6|/|D_4| = 15$ power) actually has degree 390; this is consistent with $390 \leq 405 = C_3$. □

Corollary 4. *Let* $F(q) := \det M^{(5)}[D_6] \cdot \det M^{(3)}[D_6]$ *and* $G(q) := q^{90}\big(\det M^{(4)}[D_6]\big)^2$*. Then*

$$\deg F \;\leq\; C_5{+}C_3 \;=\; 1054{+}405 \;=\; 1459, \qquad \deg G \;=\; 90{+}2\,\deg\det M^{(4)}[D_6] \;\leq\; 90{+}2{\cdot}668 \;=\; 1426,$$

and consequently $\deg(F - G) \leq 1459$*. Using the sharper value* $\deg \det M^{(3)}[D_6] = 390$ *(closed form) gives* $\deg F \leq 1054 + 390 = 1444$ *and so* $\deg(F - G) \leq 1444$*.*

3 Multi-point verification

Lemma 5. *For each integer* $q_0 \in \{2, 3, \ldots, 1450\}$ *(a set of* 1449 *distinct values),*

$$F(q_0) \;=\; G(q_0).$$

Proof. The verification script `/home/clio/projects/scratch/multi_point_verify.py` carries out, for each integer q_0 in the range, the following exact computation in $\mathbb{Z}$:

1. Build the matrices $M^{(k)}[D_6, D_6]$ at $q = q_0$, for $k \in \{3, 4, 5\}$, using direct integer Hecke right-action: each entry is a polynomial in q_0 evaluated to an integer.
2. Compute $d_k := \det M^{(k)}[D_6, D_6](q_0)$ via fraction-free Bareiss elimination (exact integer arithmetic, no rounding).
3. Compare $d_5 \cdot d_3$ with $q_0^{90} \cdot d_4^2$ as integers.

The script outputs the difference $F(q_0) - G(q_0)$ at each q_0; in every one of the 1449 runs, the difference is exactly 0. The full output (including the integer values of d_3, d_4, d_5) is logged in `/tmp/d_evals_all.txt` (concatenation of an initial single-process range $q_0 \in \{2, \ldots, 127\}$ and nine parallel chunks covering $q_0 \in \{128, \ldots, 1450\}$). □

4 Proof of Theorem 1

Proof. The polynomial $H(q) := F(q) - G(q) \in \mathbb{Z}[q]$ has degree at most 1444 by Corollary 4. By Lemma 5, $H(q_0) = 0$ for each $q_0 \in \{2, 3, \ldots, 1450\}$, a set of 1449 distinct integers. Since a non-zero polynomial of degree ≤ 1444 has at most 1444 distinct roots in any integral domain (Schwartz–Zippel/Lagrange), and $1449 > 1444$, we conclude $H \equiv 0$, i.e. $F = G$ in $\mathbb{Z}[q]$. $\square$

5 Computational reproducibility

All claims of this note are verified by exact arithmetic. The data and scripts are:

- `d5_D6_degree_analysis.py` — builds $M^{(k)}[D_6]$ as a SymPy matrix over $\mathbb{Z}[q]$ and prints per-row/per-column maximum q-degree. Output verifies Lemma 3.
- `multi_point_verify.py` — evaluates $F - G$ at each integer $q_0 \in \{2, \ldots, 1450\}$ via the integer Bareiss algorithm. Output is logged in `/tmp/d_evals_all.txt` as `(q_0, ok, d3, d4, d5, F-G)` per line; all 1449 rows have `ok` $= 1$ and $F - G = 0$. The sweep was parallelised across 9 Python processes (chunks {128–274, 275–421, ..., 1304–1450} plus an initial single-process range {2–127}) and completed in under 1 hour of wall time.
- `xmx_qdef_compute.py` — shared infrastructure: descent-paired set D_n, symbolic Hecke right-action, Rule B block decomposition.

The Bareiss step is the key ingredient. For an $m \times m$ integer matrix M, fraction-free Gaussian elimination produces $\det M$ as an exact integer in $O(m^3)$ multiplications, each of bit-length bounded by the Hadamard bound. At $q_0 = q$ and $m = 90$, the largest intermediate integer occurs near $q_0 \approx 1450$ and has fewer than 10^4 digits; Python's arbitrary precision integer arithmetic handles this without difficulty.

6 Comparison with the companion paper

The companion paper (`2026-04-19-second-q-shifted-pair.tex`) verifies the base case at only 10 primes, which establishes equality of F and G as *integer-valued functions on* 10 *specific points*, but is consistent in principle with a non-zero $F - G$ of degree as large as 1444 that happens to vanish at those specific primes. The present note closes that gap by combining the explicit degree bound (Lemma 3) with 1449 distinct evaluations.

7 Where the structural questions remain

What is *not* settled by this note is the structural question of *why* the Hirota-type identity holds. A purely algebraic proof would derive $F = G$ from a Plücker / Desnanot–Jacobi identity in some affine Grassmannian, or from the Kirillov–Reshetikhin module structure of an underlying $U_q(\widehat{\mathfrak{sl}}_2)$ representation. The current proof is honest about this: the base case is now closed by computational verification with explicit polynomial-identity rigour, but the conceptual reason remains an open question for future work.

Acknowledgements

Companion papers: `2026-04-18-first-q-shifted-pair.tex`, `2026-04-19-second-q-shifted-pair.tex`, `2026-04-18-block-decomposition-rule-b.tex`.